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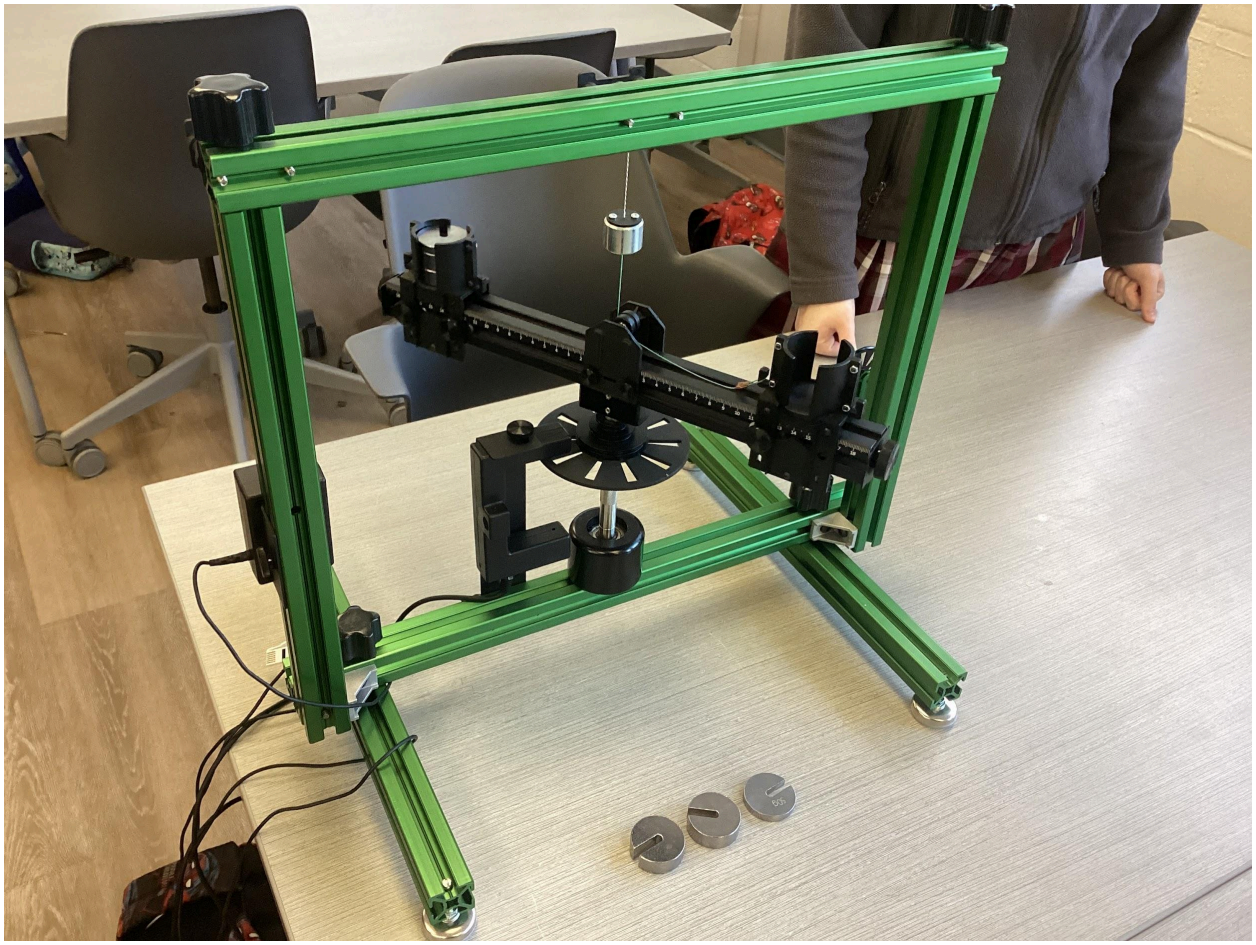
Unit 8 Lab Report

Purpose:
To determine the graphical and mathematical relationship between the mass at the extremities of the radial arm and velocity.

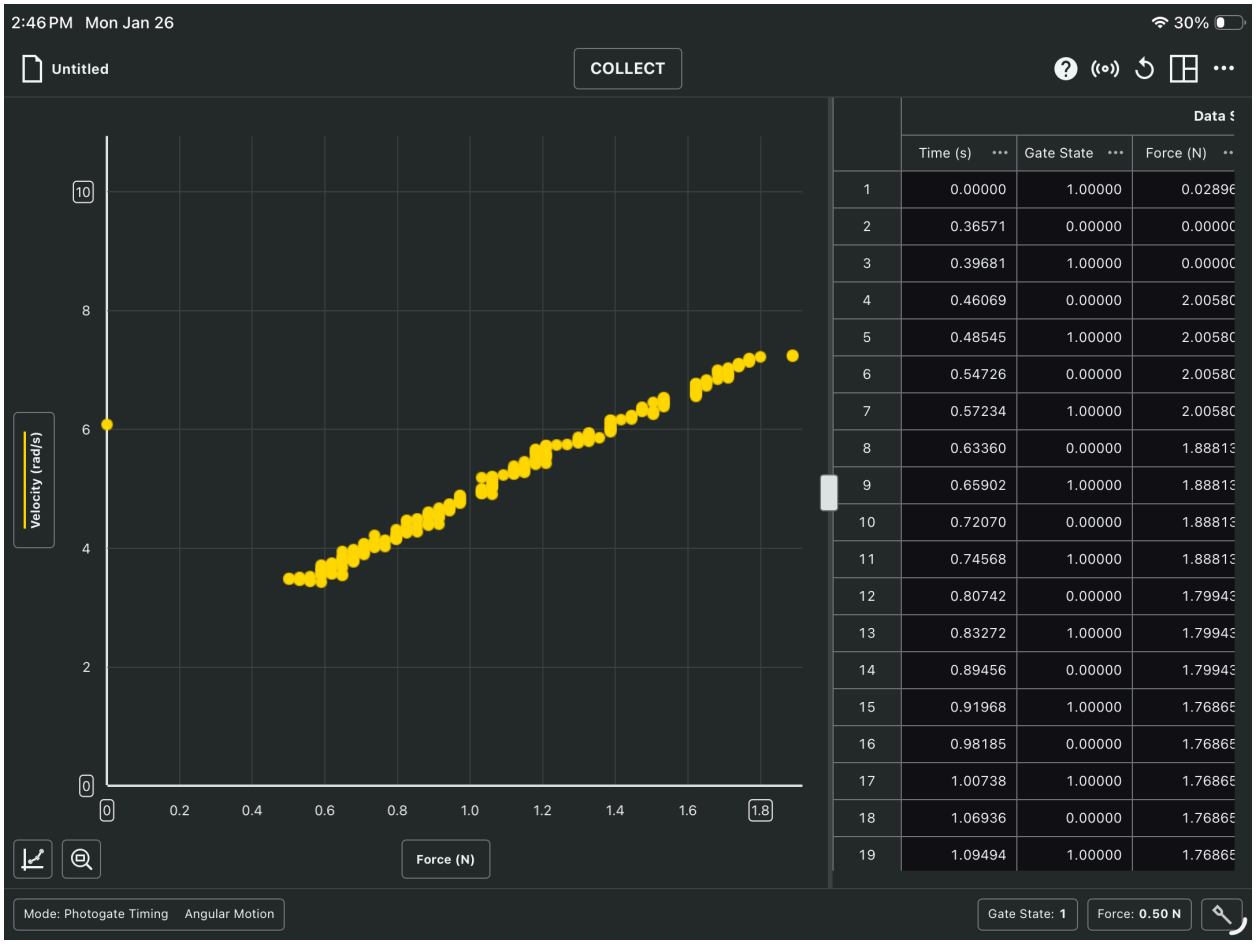
Apparatus:



Procedure:

1. The rotational inertia apparatus was placed on a flat plane, the photogate and force sensor was turned on by plugging them into the Vernier interface and connecting the interface to the Vernier Graphical Analysis with an iPad
2. The mass holders were moved to the edge of the radial arm where radius equals 16 cm.
3. Starting with no mass disk in each holder, the collect button on the Graphical Analysis App was selected, and the radial arm was spun gently.
4. Multiple seconds passed for the radial arm to begin to slow down to allow for adequate data collection.
5. Steps 3 and 4 were repeated adding a 100g disk to each mass holder for a total of 6 trials.

Data:



	Time (s)	Gate State	Force (N)
1	0.00000	1.00000	0.02896
2	0.36571	0.00000	0.00000
3	0.39681	1.00000	0.00000
4	0.46069	0.00000	2.00580
5	0.48545	1.00000	2.00580
6	0.54726	0.00000	2.00580
7	0.57234	1.00000	2.00580
8	0.63360	0.00000	1.88813
9	0.65902	1.00000	1.88813
10	0.72070	0.00000	1.88813
11	0.74568	1.00000	1.88813
12	0.80742	0.00000	1.79943
13	0.83272	1.00000	1.79943
14	0.89456	0.00000	1.79943
15	0.91968	1.00000	1.76865
16	0.98185	0.00000	1.76865
17	1.00738	1.00000	1.76865
18	1.06936	0.00000	1.76865
19	1.09494	1.00000	1.76865

-We found the angular velocity in rad/s by interpolating the graph to estimate the angular velocity when the force was 0.60N.

mass (g)	Velocity (rad/s)
0	8
100	6
200	5.3
300	4.3
400	3.7
500	3.5

Analysis:

1/m (1/kg)	Velocity^2 (m^2/s^2)
Not measured	1.638

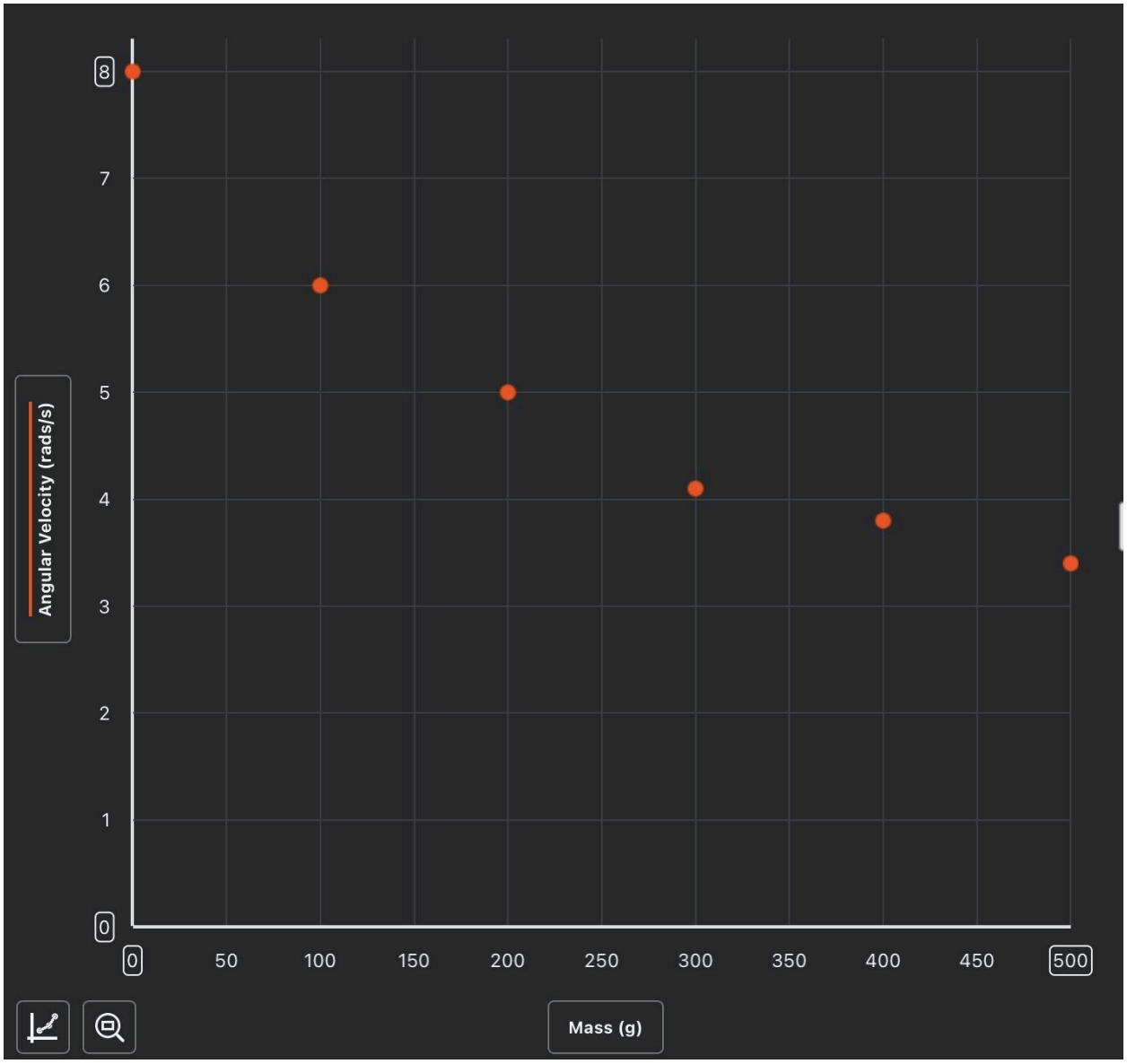
10.0	.922
5.0	.640
3.33	.430
2.50	.370
2.0	.296

Masses were converted from grams to kilograms by dividing by 1000. Angular velocity was multiplied by the constant radius of .16m to find the linear velocity in meters per second.

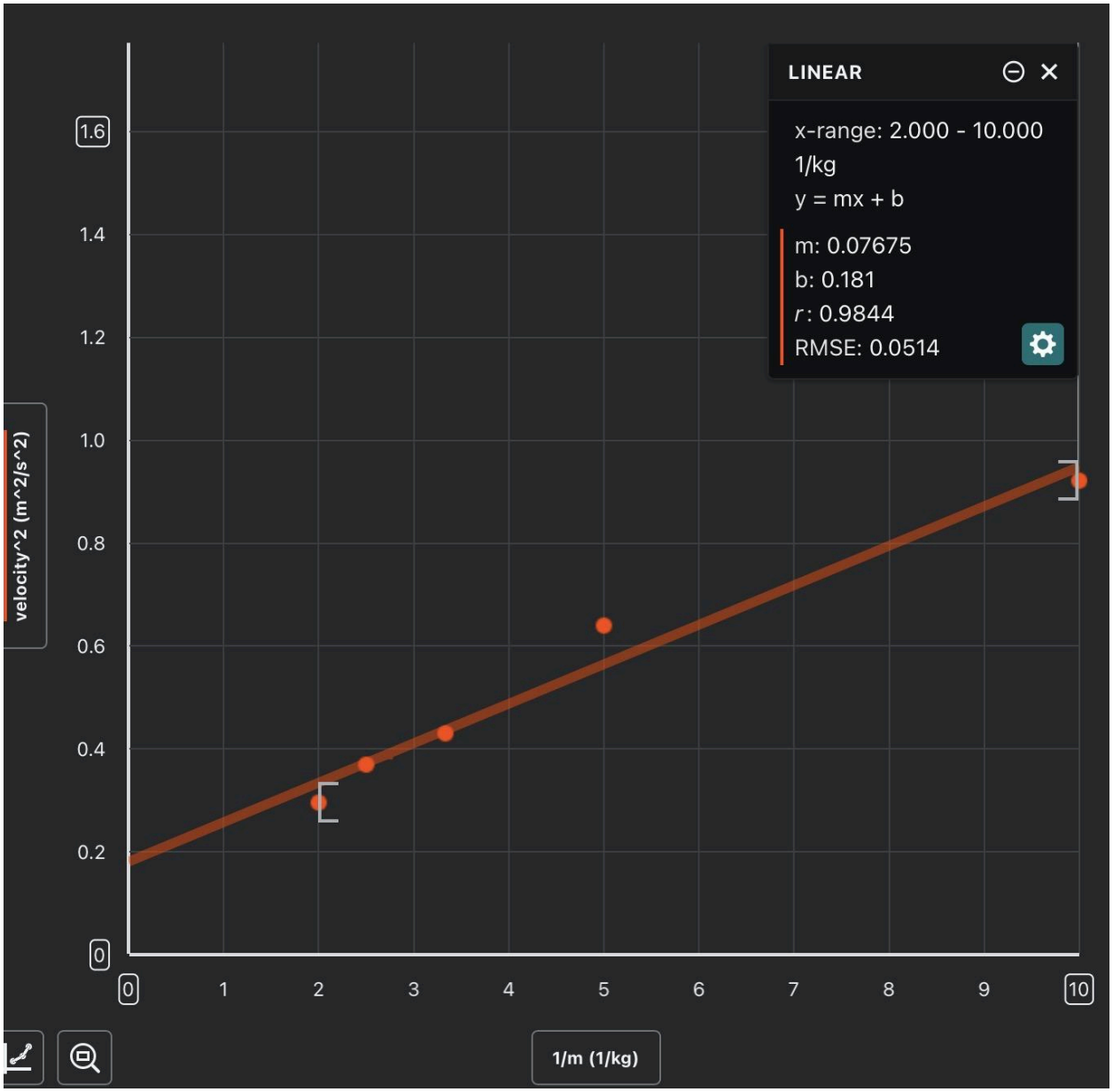
Sample Calculations:

Mass conversion: m = 100g/1000=0.1 kg
Linear velocity: v= ωr = 6 rad/s * 0.16m = 0.96 m/s

Graph of original data:



Linearized graph:



v^2 m^2/s^2 = (0.077 kg m^2/s^2) x + 0.181 m^2/s^2

Conclusion:

In the original data, an inversely proportional relationship was found between mass and velocity. To determine the mathematical relationship, the angular velocity values were first converted into linear velocity as shown in the sample calculations above. When velocity was graphed against mass, the result was the curved graph that needed to be linearized. To linearize the graph, velocity was squared and the inverse of mass was taken. These transformed variables then created a straight-line graph. Utilizing the Graphical Analysis App, a curve fit was drawn. The slope of the line and y-intercept were used to write an equation in the form y=mx+b. This resulted in the equation found below:

$$v^2 = \frac{0.077}{m}\left(\frac{kg \cdot m^2}{s^2}\right) + 0.181\left(\frac{m^2}{s^2}\right)$$

We observed that as 1/m increases, v^2 increases in a linear fashion. The curve fit line had a small y-intercept, showing that the relationship is not perfectly proportional, likely due to experimental error. However, the data still closely follows the expected inverse relationship:

$$v^2 \propto \frac{1}{m}$$

The slope of the found equation represents a constant that comes from the experimental setup. In this particular experiment, the slope of the graph (0.077) represents the constant of proportionality. In the experiment, the radius of 0.16m and the centripetal force of 0.60N were held constant. The slope corresponds to the product of these constants. The general equations for centripetal force is:

$$F_{net} = \frac{mv^2}{r}$$

Rearranging this equation to solve for v^2 results in the generalized equation below.

$$v^2 = \frac{Fr}{m}$$

This matches the form of the experimental equation, where the slope (0.077) is equivalent to the product of Fr. To verify this, units can be compared. The left side of the equation v^2 has units of m^2/s^2. On the right side of the equation, the slope is multiplied by 1/m resulting in the slope units of kgm^2/s^2. This corresponds to the multiplication of our constant Fr:

$$Fr = (\frac{kgm}{s^2})(m) = \frac{kgm^2}{s^2}$$

When an object is moving in a circle, it needs to constantly change direction. This is why the speed of an object moving around a circle can be consistent but the velocity will be changing (velocity includes direction). Therefore, there is a net force acting on the object which always points towards the center of the circle in order to keep the object on path. This force is called the centripetal force. The following equation describes the net inward force needed to keep the object moving in its path around the circle, where m=mass of the object, v=speed of the object, and r=radius of the object.

$$F_{net} = \frac{mv^2}{r}$$

Newton's first law describes that an object will continue moving in a straight line at a constant velocity until a net force acts on it. Therefore, it is natural for an object in circular motion to want to move in a straight line rather than a circular path. However, this doesn't happen because the centripetal force pulls the object inward to continue its circular motion and avoid the object leaving to go in a straight path. Therefore, circular motion shows an exception of Newton's first law, which can be attributed only to the net force acting on the object. Newton's second law states Fnet=ma, but in circular motion a=v^2/r. If this acceleration for circular motion were to be plugged into Newton's second law, we would get Fnet=mv^2/r (shown above). This demonstrates that the centripetal force is just the net force needed for inward acceleration in circular motion. Therefore, we can recognize that centripetal force isn't a new type of force, but rather is Newton's second law being applied to a curved path.

One source of error towards the beginning of this procedure could have been variations between how the radial arm was spun between each trial, which could've impacted the initial angular velocity. To correct this, we can ensure we are always moving the radial arm in the same direction to avoid different amounts of friction. Additionally, we can use practice runs with feedback from Graphical Analysis App to aim for more consistent and clustered initial angular velocities (as opposed to random initial angular velocities). Another source of error could've been that the "collect data" button was clicked at various relative times across trials of the experiment. To correct this, we can create a system where the experimenter who spins the radial arm says "spin" to signal the other experimenter to click the button before they actually spin. This could ensure that the button is being pressed almost at the same amount of time before the spinning of the radial arm throughout trials. A third source of error could've been the data collection window changing between trials based on how long it took the radial arm to stop spinning. To correct this, we could choose, prior to the experiment, a certain amount of time for data to be collected on the Graphical Analysis App, which would remain consistent. The data collection window of time would decrease variations and timing errors across trials.